

第三章 干涉

§1. 波的干涉与叠加

1. 波的描述

$$u(r, t) = A(r) \cos[\omega t - \varphi(r)]$$

$$\Leftrightarrow u(r, t) = A(r) e^{-i[\omega t - \varphi(r)]} \quad \begin{array}{l} \varphi(r) + \Rightarrow \text{超前} \\ - \Rightarrow \text{落后} \end{array}$$

$$= \tilde{u}(r) e^{-i\omega t}$$

$$\text{复振幅 } \tilde{u}(r) = A(r) e^{i\varphi(r)}$$

$$\text{光强: } I(r) = [A(r)]^2 = u(r) \tilde{u}(r)$$

2. 波的叠加 (线性光学)

$$u(r, t) = \sum_i u_i(r, t)$$

对 ω 相同的同频波:

$$\tilde{u}(r, t) = \sum_i \tilde{u}_i(r, t)$$

3. 两光源干涉干涉

$$I(r) = [A(r)]^2 = \tilde{u}(r) \tilde{u}^*(r)$$

$$= [\tilde{u}_1(r) + \tilde{u}_2(r)] [\tilde{u}_1^*(r) + \tilde{u}_2^*(r)]$$

$$= A_1^2(r) + A_2^2(r) + \tilde{u}_1 \tilde{u}_2^* + \tilde{u}_2 \tilde{u}_1^*$$

$$= A_1^2 + A_2^2 + A_1 A_2 [e^{i(\varphi_1 - \varphi_2)} + e^{-i(\varphi_1 - \varphi_2)}]$$

$$= A_1^2 + A_2^2 + 2 A_1 A_2 \cos(\varphi_1 - \varphi_2)$$

$$= A_1^2 + A_2^2 + 2 A_1 A_2 \cos \delta \varphi$$

取 $A_1 = A_2 = A$ (远场条件下, $r_1, r_2 \gg d$) $d \cdot$ r_1
 r_2

$$I(P) = A^2 [2 + 2 \cos \delta] = 4A^2 \cos^2(\frac{\delta}{2})$$

$$\varphi_1 = \varphi_{10} + 2\pi \cdot \frac{r_1}{\lambda}$$

$$\varphi_2 = \varphi_{20} + 2\pi \cdot \frac{r_2}{\lambda}$$

$$\delta(P) = \varphi_1 - \varphi_2 = \underbrace{(\varphi_{10} - \varphi_{20})}_{=0} + \frac{2\pi}{\lambda} \cdot (r_1 - r_2)$$

$$= \frac{2\pi}{\lambda} \cdot \Delta L = \begin{cases} 2k\lambda & \text{极大} \\ (2k+1)\lambda & \text{极小} \end{cases}$$

$$\Rightarrow \Delta L = \begin{cases} k\lambda & \text{极大} \\ (k + \frac{1}{2})\lambda & \text{极小} \end{cases}$$

§ 2. 杨氏双缝 空间相干性

1. Young 双缝干涉实验



$$\varphi_{10} = \varphi_0 + 2\pi \cdot \frac{r_1}{\lambda}$$

$$\varphi_{20} = \varphi_0 + 2\pi \cdot \frac{r_2}{\lambda}$$

$$r_1, L \gg d, \Delta x$$

$$r_2 = \sqrt{L^2 + (x + \frac{d}{2})^2} \approx L + \frac{1}{2L} (x + \frac{d}{2})^2$$

$$\Delta L = r_1 - r_2 = \frac{1}{2L} [(x - \frac{d}{2})^2 - (x + \frac{d}{2})^2] = \frac{1}{2L} (-d)(2x) = -\frac{xd}{L}$$

$$\delta = \frac{2\pi}{\lambda} \cdot \Delta L = - \frac{2\pi d}{\lambda L} \cdot x$$

$$\text{取 } \delta = 2\pi \rightarrow \text{条纹间隔 } \Delta x = \frac{\lambda L}{d}$$

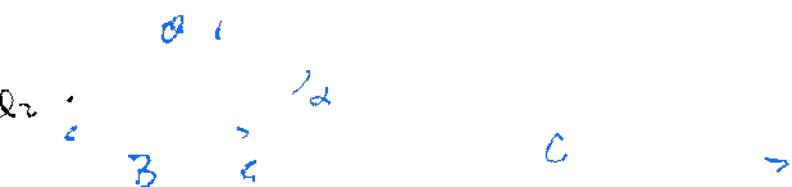
2. 干涉装置

(1) Fresnel 镜

• a

Q1.

$$\Delta x = \frac{\lambda(B+C)}{2\alpha B}$$



S1 - a

$$\Delta x = \frac{\lambda(B+C)}{2(n-1)\alpha B}$$



(2) Lloyd 镜



$$\Delta x = \frac{\lambda D}{2\alpha}$$



3. 干涉条纹的移动

光源位置移动 δs 时,

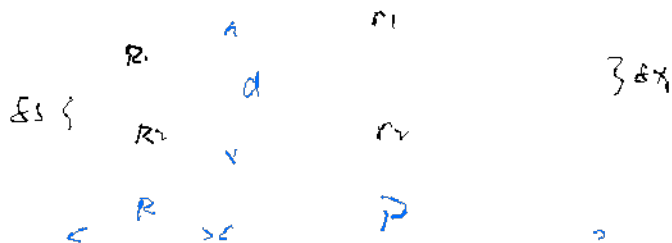
同级干涉条纹移动 δx

$$\Delta L = R_1 + r_1 - R_2 - r_2$$

$$= (R_1 - R_2) + (r_1 - r_2)$$

$$R_1 - R_2 = d \cdot \frac{\delta s}{R} \quad r_1 - r_2 = -d \cdot \frac{\delta x}{D}$$

$$\text{对同一级条纹 } \Delta L = 0 \Rightarrow \delta x = -\frac{D}{R} \delta s$$



4. 微观发光机制

持续时间 $\tau < 10^{-15} \text{ s} \Rightarrow L < m$

且振动方向、相位均随机 $\Rightarrow \overline{\cos(\varphi_1 - \varphi_2)} = 0$

$\Rightarrow$ 非相干叠加, $I(P) = I_1(P) + I_2(P)$

5. 光源宽度对对比度的影响

$$\sigma = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

对于非理想狭缝光源 (有宽度的)

$$\delta x = \Delta x \text{ 时 } \sigma = 0 \Rightarrow \text{光源极限亮度 } b_0 = \frac{R}{D} \frac{\Delta D}{\Delta x} = \frac{R}{D}$$

$$I(x) = I_0 (1 + \cos \Delta \pi)$$

$$= I_0 \left[1 + \cos \left(\frac{2\pi d}{\lambda D} x \right) \right] = I_0 \left[1 + \cos \left(2\pi \cdot \frac{x}{\lambda x} \right) \right]$$

当光源移动 s , 条纹移动 $\delta x = -\frac{D}{R} s$

$$I(x, s) = I_0 \left[1 + \cos \frac{2\pi d}{\lambda D} (x + \delta x) \right] = I_0 \left[1 + \cos \left(\frac{2\pi d}{\lambda D} x - \frac{2\pi d}{\lambda R} s \right) \right]$$

$$= I_0 \left[1 + \cos \frac{2\pi d}{\lambda D} x \cos \left(\frac{2\pi d}{\lambda R} s \right) + \sin \left(\frac{2\pi d}{\lambda D} x \right) \sin \left(\frac{2\pi d}{\lambda R} s \right) \right]$$

$$I(x) = \frac{1}{b} \int_{-\frac{b}{2}}^{\frac{b}{2}} I(x, s) ds$$

$$= \frac{I_0}{b} \int_{-\frac{b}{2}}^{\frac{b}{2}} \left[1 + \cos\left(\frac{2\pi a}{\lambda D} x\right) \cos\left(\frac{2\pi a}{\lambda R} s\right) + \underbrace{\sin\left(\frac{2\pi a}{\lambda D} x\right) \sin\left(\frac{2\pi a}{\lambda R} s\right)}_{\text{奇函数积分为0}} \right] ds$$

$$= \frac{I_0}{b} \int_{-\frac{b}{2}}^{\frac{b}{2}} \left[1 + \cos\left(\frac{2\pi a}{\lambda D} x\right) \cos\left(\frac{2\pi a}{\lambda R} s\right) \right] ds$$

$$= \frac{I_0}{b} \left\{ b + \cos\left(\frac{2\pi a}{\lambda D} x\right) \frac{\lambda R}{2\pi a} \cdot \left[\sin\left(\frac{\pi a b}{\lambda R} \right) - \sin\left(-\frac{\pi a b}{\lambda R} \right) \right] \right\}$$

$$= I_0 \left\{ 1 + \frac{\sin \frac{\pi a b}{\lambda R}}{\frac{\pi a b}{\lambda R}} \cdot \cos\left(\frac{2\pi a}{\lambda D} x\right) \right\} \quad \text{记 } u = \frac{\pi a b}{\lambda R}$$

$$= I_0 \left[1 + \frac{\sin u}{u} \cos\left(\frac{2\pi a}{\lambda D} x\right) \right] = I_0 \left(1 + \sin u \cos \frac{2\pi a}{\lambda D} x \right) \quad \sin u = \frac{\sin u}{u} \cdot u$$

$$I_{\max} = I_0 (1 + \sin u) \quad I_{\min} = I_0 (1 - \sin u)$$

$$\gamma = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = |\sin u|$$

6. 空间相干性

最大亮度: $\delta x = -\frac{D}{R} \delta s \rightarrow b_0 = \frac{\lambda R}{d}$

($b > b_0$, $\sigma = 0$, A, B 不相干)

给定亮度为 b 的白光, 多靠近 S_1, S_2 是 (部分) 相干的?

定义: 相干程度: $\alpha = \frac{R \lambda}{b}$

孔径角: $\Delta \theta_0 = \frac{d}{R} = \frac{\lambda}{b}$

or $\alpha \approx \Delta \theta_0$

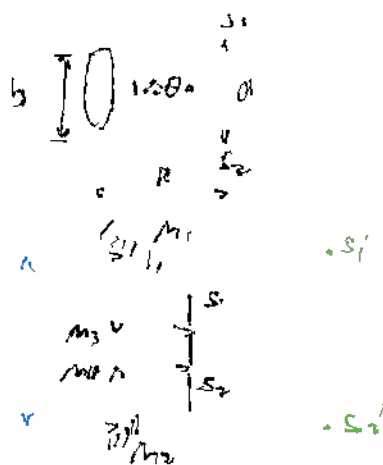
例: 迈克尔逊干涉仪 (测量星体角直径)

调大 b , 直至干涉条纹消失, 求星体角直径

S_1', S_2' 为 S_1, S_2 的像, 两者空间相干程度相同

$\Delta \theta_0 = \frac{\lambda}{b} = \frac{d}{R} \rightarrow \frac{R}{R} = \frac{\lambda}{b}$

角直径 $\varphi = \frac{b}{R} = \frac{\lambda}{d}$ (取 $\lambda = 5.55 \times 10^{-7} \text{ m}$)



§ 3. 等厚条纹

1. 振幅干涉

2. $AB \approx BC$?

$$\Delta L \textcircled{2} \frac{2nh}{\cos i} - 2nh \tan i \cdot \sin i$$

$$n \sin i_1 = n \sin i_2$$

$$= 2nh \left(\frac{1}{\cos i} - \frac{\sin^2 i}{\cos i} \right) = 2nh \cos i$$

$$\Delta L = \begin{cases} k\lambda \\ (k+\frac{1}{2})\lambda \end{cases} \rightarrow h = \begin{cases} \frac{k\lambda}{2n \cos i} \\ \frac{(2k+1)\lambda}{4n \cos i} \end{cases}$$



★ (反射) 半波损失:

$$\begin{cases} n_1 < n > n_2, & n_1 > n < n_2 \rightarrow \text{有} \\ n_1 > n > n_2, & n_1 < n < n_2 \rightarrow \text{无} \end{cases}$$

$$\delta = 0 \rightarrow \Delta L = 2nh \quad \Delta h = \frac{\lambda}{2n}$$

3. $\frac{\lambda}{2}$ 的倍

$$\Delta h = d \Delta x \rightarrow \Delta x = \frac{\lambda}{2nd}$$

Q: 如何判断条纹位置?

A: 按压一侧, 观察 Δx 变化

4. 牛顿环

$$\begin{aligned}\Delta L_k &= 2 \cdot (R - \sqrt{R^2 - r_k^2}) \\ &\approx 2 \left[R - R \left(1 - \frac{r_k^2}{2R^2} \right) \right] \\ &= \frac{r_k^2}{R} = k\lambda\end{aligned}$$

$$\rightarrow r_k = \sqrt{k \cdot R \cdot \lambda}$$

消除 0 处不密接误差?

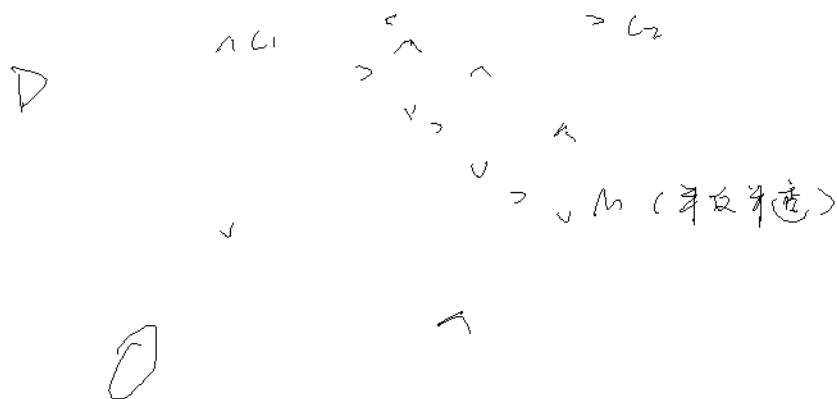
$$\hookrightarrow \Delta L = d_0 + \frac{r_k^2}{R} = k\lambda$$

做差: 则第 m, n 级条纹

$$\begin{cases} d_0 + \frac{r_m^2}{R} = m\lambda \\ d_0 + \frac{r_n^2}{R} = n\lambda \end{cases} \rightarrow R = \frac{(r_m^2 - r_n^2)}{(m - n)\lambda}$$

5. 等厚条纹观测, 衬比度

精密:



影响:

① 条纹形状: 偏离直线

$$\Delta L = 2nh \cos i$$

$$d(\Delta L) = 2n \cos i dh - 2nh \sin i di = 0$$

$$\Rightarrow dh = h \tan i di \quad \text{即 } \Delta L = 0 \text{ 时 } h \neq i$$

$$\ln h = \int - \frac{d(\cos i)}{\cos i} = -\ln(\cos i) + C$$

$$h = \frac{C'}{\cos i}$$

$$\Rightarrow \Delta L = \text{const.} \quad h \neq i$$

$\Rightarrow$ 条纹弯曲交横

② 对比度下降

$\uparrow$ 光源亮度

(图像: 等倾线^①与等厚线^②交点)

解析: 严格?

□ 每过一个周期 ΔL 向外减少 λ

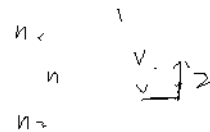
□ 每过一个周期 ΔL 向右增大 λ

$$\Delta L_{p1} - \Delta L_{p2} = \lambda - \lambda = 0$$

6. 薄膜颜色 增透膜

低膜 L: $n < n_2$ ($n > n_1$, 无半波损失)

高膜 H: $n > n_2$ ($n > n_1$, 有半波损失)



相膜: $2nh = \frac{\lambda}{2} \Rightarrow \Delta \varphi = \pi$, 1, 2 相消

$n = \sqrt{n_1 n_2}$ 时, 完全相消

相机 $\rightarrow$ 消 550 nm 光的反射 $\rightarrow$ 呈蓝紫色

§ 4 等倾条纹

$$1. \Delta L = 2nh \cos i$$

$$\Delta L = \begin{cases} k\lambda & \text{亮} \\ (k + \frac{1}{2})\lambda & \text{暗} \end{cases} \Rightarrow \cos i_k \begin{cases} \frac{k\lambda}{2nh} & \text{亮} \\ \frac{(2k+1)\lambda}{4nh} & \text{暗} \end{cases}$$

$$\Delta(\cos i_k) = \frac{\lambda}{2nh}$$

$$\frac{d(\cos i)}{di} \cdot (i_{k+1} - i_k) = (i_{k+1} - i_k)$$

光源
a

$$= -\sin i_k (i_{k+1} - i_k)$$

$$\Rightarrow i_{k+1} - i_k = -\frac{\lambda}{2nh \sin i_k}$$

当 i_k 较小, $r \propto i$

$$\therefore \Delta r = r_{k+1} - r_k \propto -\frac{\lambda}{2nh \sin i_k}$$

$$\Rightarrow i_k \nearrow \Delta r \downarrow$$

$$h \nearrow \Delta r \downarrow$$

* 使用扩展光源不影响干涉度

说明: 到达幕上同一点的光必定以相同的角度进入透镜 L

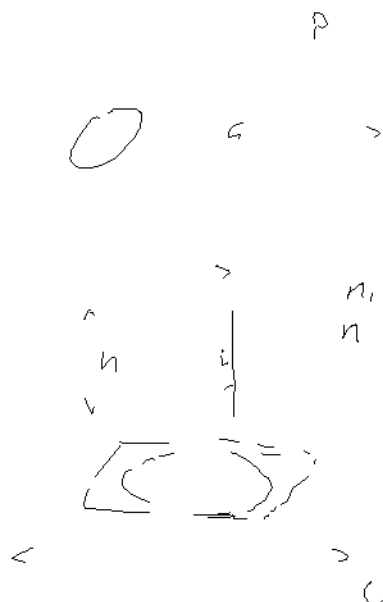
因此 ΔL 相同

3. 定域问题

b.c., a.d.

P ← 定域中心层

$$b \Delta \theta < \lambda$$



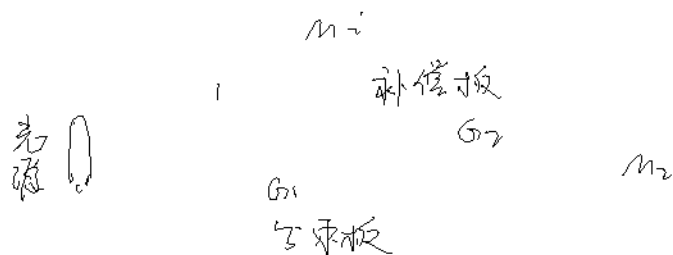
厚度均匀薄膜 $\theta = 0$

可以使用扩展光源

n_1
 n
 n_2

§ 5. 迈克耳孙干涉仪 时间相干性

1. 装置结构 n_1



2. 干涉条纹

1) 等倾

呈圆形

2) 等厚

))) / ||||| / |||||

判断中心0级条纹:

$\theta = 0$ 且有半波损 $\rightarrow$ 暗纹

白光照明, 加入补偿板 G_2 使0级无色散

5.3 非单色性

$$\frac{dZ_{\lambda}}{d\lambda} \propto$$

为双缝结构 (离散)

$$I_1 = I_0 [1 + \cos(k_1 \Delta L)] \quad k_1 = \frac{2\pi}{\lambda_1}$$

$$I_2 = I_0 [1 + \cos(k_2 \Delta L)] \quad k_2 = \frac{2\pi}{\lambda_2}$$

$$\lambda_1 \approx \lambda_2 \approx \lambda$$

$$I = I_1 + I_2 = I_0 [2 + \cos(k_1 \Delta L) + \cos(k_2 \Delta L)]$$

$$= I_0 [2 + 2 \cos\left(\frac{k_1 + k_2}{2} \Delta L\right) \cos\left(\frac{k_1 - k_2}{2} \Delta L\right)]$$

其中: $\Delta k = k_1 - k_2 \approx \frac{k_1 + k_2}{2} \Rightarrow$ 近似认为 $\cos\left(\frac{k_1 - k_2}{2} \Delta L\right) = \cos 0 = 1$

$$= 2I_0 [1 + \cos\left(\frac{k_1 + k_2}{2} \Delta L\right)]$$

$$\sigma = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{1 + \cos\left(\frac{\Delta k}{2} \Delta L\right) - [1 - \cos\left(\frac{\Delta k}{2} \Delta L\right)]}{1 + \cos\left(\frac{\Delta k}{2} \Delta L\right) + 1 - \cos\left(\frac{\Delta k}{2} \Delta L\right)}$$

$$= |\cos\left(\frac{\Delta k}{2} \Delta L\right)|$$

$$\rightarrow \sigma = 0 \Rightarrow \frac{\Delta k}{2} \Delta L = \frac{\pi}{2}$$

$$\text{即 } \Delta L = \frac{\pi}{\Delta k} = \frac{\pi}{k_1 - k_2} = \frac{\lambda_1 \lambda_2}{2(\lambda_2 - \lambda_1)}$$

$$\approx \frac{\lambda^2}{2\Delta\lambda}$$

当 $\lambda_1 < \lambda_2$ 时, λ_1 的 k 级谱纹与 λ_2 的 k 级谱纹重合时 $\sigma = 0$

解释:

$$\Delta L = (k + \frac{1}{2}) \lambda_1 \quad \text{暗} \quad \rightarrow k = \frac{\lambda_1}{2(\lambda_2 - \lambda_1)}$$

$$= k \lambda_2 \quad \text{亮}$$

$$\Delta L = \frac{\lambda_1 \lambda_2}{2(\lambda_2 - \lambda_1)} \approx \frac{\lambda^2}{2\Delta\lambda}$$

当 $\lambda_1 < \lambda_2$ 时, λ_1 的 k 级谱纹与 λ_2 的 k 级谱纹重合时 $\sigma = 0$

光栅-光栅 → 衍射 : $\Delta L = N_1 \lambda_1 = N_2 \lambda_2 = (N_1 - \frac{1}{2}) \lambda_2$

$$\Rightarrow N_1 = \frac{\lambda_2}{2\lambda_1 - \lambda_2} \approx \frac{\lambda_2}{2\Delta\lambda}$$

再动 N_1 , 恢复原状 → 空间周期 $2N_1\lambda_1$, 频率 $\frac{1}{2N_1\lambda_1} \approx \frac{\Delta\lambda}{\lambda^2} = \frac{\Delta f}{2\lambda}$

单色线宽 (连续)

引光谱密度 $i(\lambda) = \frac{dI}{d\lambda}$ $I = \int_0^\infty i(\lambda) d\lambda = \text{①} \int_0^\infty i(k) dk$
约化系数

$$I(\Delta L) = \frac{1}{\lambda} \int_0^\infty i(k) [1 + \cos(k\Delta L)] dk$$

$$= I_0 + \frac{1}{\lambda} \int_0^\infty i(k) \cos(k\Delta L) dk$$

选取矩形 $i(k) = \begin{cases} \frac{I_0}{\Delta k} & k \in (k_0 - \frac{\Delta k}{2}, k_0 + \frac{\Delta k}{2}) \\ 0 & k \notin (k_0 - \frac{\Delta k}{2}, k_0 + \frac{\Delta k}{2}) \end{cases}$ 满足归一化条件 $\frac{1}{\lambda} \int_0^\infty i(k) dk = I_0$

$$\text{则 } I(\Delta L) = I_0 + \frac{1}{\lambda} \int_{k_0 - \frac{\Delta k}{2}}^{k_0 + \frac{\Delta k}{2}} \frac{I_0}{\Delta k} \cos(k\Delta L) dk$$

$$= I_0 \left[1 + \frac{1}{\Delta k} \int_{k_0 - \frac{\Delta k}{2}}^{k_0 + \frac{\Delta k}{2}} \cos(k\Delta L) dk \right]$$

$$= I_0 \left[1 + \frac{\sin(k_0 + \frac{\Delta k}{2})\Delta L - \sin(k_0 - \frac{\Delta k}{2})\Delta L}{\Delta k \cdot \Delta L} \right]$$

$$= I_0 \left[1 + 2 \cos(k_0\Delta L) \sin(\frac{\Delta k \Delta L}{2}) \right]$$

$$= I_0 \left[1 + \cos(k_0\Delta L) \cdot \text{sinc}(\frac{\Delta k \Delta L}{2}) \right] \quad \Delta k \ll k_0$$

$$\alpha \left[\text{sinc} \frac{\Delta k \Delta L}{2} \right] \approx 0$$

$$\Rightarrow \delta = \left| \text{sinc} \left(\frac{\Delta k \Delta L}{2} \right) \right|$$

$$\delta = 0 \Rightarrow \Delta L_{\max} \frac{2\pi}{\Delta k} \approx \frac{\lambda^2}{\Delta\lambda}$$

" $\lambda_1 < \lambda_2$ 时, λ_1 的 $k+1$ 级亮纹与 λ_2 的 k 级亮纹重合"

解释: $\Delta L = (k+1)\lambda_1$ $k = \frac{\lambda_1}{\lambda_2 - \lambda_1}$

$$= k\lambda_2$$

$$\Delta L = \frac{\lambda_1 \lambda_2}{\lambda_2 - \lambda_1} \approx \frac{\lambda^2}{\Delta \lambda}$$

4. 傅里叶变换光谱仪

已知 $I(\Delta L)$, 求 $i(k)$

余弦傅里叶变换:

$$I(\Delta L) - I_0 = \frac{1}{\lambda} \int_0^{\infty} i(k) \cos(k \Delta L) dk$$

逆变换:

$$i(k) = 2 \int_0^{\infty} [I(\Delta L) - I_0] \cos(k \Delta L) d(\Delta L)$$

光谱仪: $\Delta L = 2vt$ $\lambda = vt$

$$i(\Delta L) \rightarrow i(t) \rightarrow I(t) \rightarrow I(x)$$

6. 时间相干性

相干光: 波列长度 $l_0 = v \cdot \tau_0$

$$光程 L_0 = n l_0 = c \cdot \tau_0$$

对于点 P_1, P_2 $\Delta L \begin{cases} > L_0 & \text{不相干} \\ < L_0 & \text{部分相干} \\ = 0 & \text{完全相干} \end{cases}$

L_0 : 相干长度 τ_0 : 相干时间

与 ΔL_{max} 同量级

沿 x 方向传播的平面波

$$\text{振幅 } \vec{u} = \vec{u} e^{ikx} \quad \text{定义 } \vec{a}(k) = \frac{d\vec{u}}{dk}$$

$$= \frac{1}{\pi} \int_0^{\infty} \vec{a}(k) e^{ikx} dk$$

使用 δ 函数模型 $\tilde{A}(k) = \begin{cases} \frac{\tilde{A}}{\Delta k} & k \in (k_0 - \frac{\Delta k}{2}, k_0 + \frac{\Delta k}{2}) \\ 0 & k \notin (k_0 - \frac{\Delta k}{2}, k_0 + \frac{\Delta k}{2}) \end{cases}$

$$\Rightarrow \tilde{u} = \frac{1}{\lambda} \int_{k_0 - \frac{\Delta k}{2}}^{k_0 + \frac{\Delta k}{2}} \frac{\tilde{A}}{\Delta k} e^{ikx} dk = \frac{\tilde{A}}{\Delta k} \int_{k_0 - \frac{\Delta k}{2}}^{k_0 + \frac{\Delta k}{2}} e^{ikx} dk$$

$$= \frac{\tilde{A}}{\Delta k} \frac{1}{ix} [e^{i(k_0 + \frac{\Delta k}{2})x} - e^{i(k_0 - \frac{\Delta k}{2})x}]$$

$$= \frac{\tilde{A}}{\Delta k} \frac{e^{ik_0 x}}{ix} (e^{\frac{1}{2}i\Delta k x} - e^{-\frac{1}{2}i\Delta k x})$$

$$= \tilde{A} e^{ik_0 x} \frac{2 \sin(\frac{1}{2}\Delta k x)}{\Delta k x} = \tilde{A} \cdot \text{sinc}(\frac{1}{2}\Delta k x) e^{ik_0 x}$$

振幅 $\tilde{A} \text{sinc}(\frac{1}{2}\Delta k x) = 0$

$$\Rightarrow \frac{1}{2}\Delta k (\frac{1}{2}L_0) = \pi \Rightarrow \frac{1}{2}L_{\max} = \frac{2\pi}{\Delta k} \approx \frac{\lambda^2}{\Delta \lambda}$$

$$\lambda \times \Delta \lambda = \Delta \frac{c}{\lambda} \approx \frac{c \Delta \lambda}{\lambda^2}$$

$$L_0 \Delta \lambda \approx 1$$

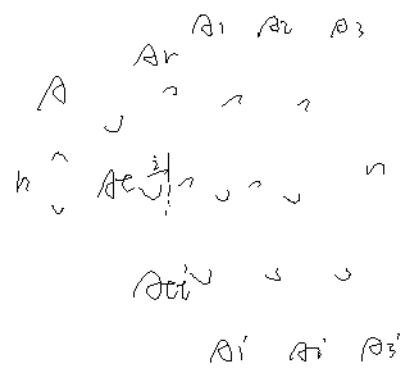
§ 6. 多光束干涉 F-P干涉仪

1. 多光束干涉

r, t 膜外 $\rightarrow$ 膜内 的 反、透射率

r', t' 膜内 $\rightarrow$ 膜外 的 反、透射率

stokes 关系: $r = -r'$, $r^2 + t t' = 1$
 相位相反



$A_1 = A r$	$A_1' = A t t'$
$A_2 = A t r' t'$	$A_2' = A t r'^2 t'$
$A_3 = A t r'^3 t'$	$A_3' = A t r'^4 t'$
$\vdots$	$\vdots$

$\Delta L = 2 n h \cos i$. $\delta = k \Delta L = \frac{4 \pi n h \cos i}{\lambda}$

$\tilde{u}_1 = -A r'$	$\tilde{u}_1' = A t t'$
$\tilde{u}_2 = A t r' t' e^{i \delta}$	$\tilde{u}_2' = A t r'^2 t' e^{i \delta}$
$\tilde{u}_3 = A t r'^3 t' e^{2 i \delta}$	$\tilde{u}_3' = A t r'^4 t' e^{2 i \delta}$

$$\begin{aligned} \tilde{u}_R = \sum \tilde{u}_j &= A [-r' + t t' r' e^{i \delta} (1 + r'^2 e^{i \delta} + \dots + \dots)] \\ &= A [-r' + t t' r' e^{i \delta} \frac{1}{1 - r'^2 e^{i \delta}}] \end{aligned}$$

$$= A r' \left(\frac{1 + e^{i\delta}}{1 - r'^2 e^{i\delta}} - 1 \right) = r' \frac{(1 + e^{i\delta})e^{i\delta} - 1}{1 - r'^2 e^{i\delta}} A$$

$$= \frac{r'(e^{i\delta} - 1)}{1 - r'^2 e^{i\delta}} A = \frac{r'(e^{i\delta} - 1)(1 - r'^2 e^{-i\delta})}{(1 - r'^2 e^{i\delta})(1 - r'^2 e^{-i\delta})} A$$

$$= \frac{r'(-1 - r'^2 + e^{i\delta} + r'^2 e^{-i\delta})}{1 + r'^4 - 2r'^2 \cos \delta} = \frac{r'[-(1 + r') + 2r'^2 \cos \delta] + (1 - r'^2)e^{i\delta}}{r'^4 - 2r'^2 \cos \delta + 1}$$

$$I_R = \tilde{U}_R \cdot \tilde{U}_R^* = r'^2 [2r'^2 \cos \delta - 1 - r' + (1 - r'^2)e^{i\delta}] \cdot [2r'^2 \cos \delta - 1 - r' + (1 - r'^2)e^{-i\delta}]$$

$$(r'^4 - 2r'^2 \cos \delta + 1)^{-1}$$

$$= \frac{r'^2 (2r'^2 \cos \delta - 1 - r')^2 + (1 - r'^2)^2 + 2(2r'^2 \cos \delta - 1 - r')(1 - r'^2) \cos \delta}{(r'^4 - 2r'^2 \cos \delta + 1)^{-1}}$$

= - -

$$\tilde{U}_T = \sum U_j' = A t t' (1 + r'^2 e^{i\delta} + \dots + \dots)$$

$$= A t t' \frac{1}{1 - r'^2 e^{i\delta}}$$

$$I_T = \tilde{U}_T \cdot \tilde{U}_T^* = \frac{(A t t')^2}{(1 - r'^2 e^{i\delta})(1 - r'^2 e^{-i\delta})} = \frac{(A t t')^2}{1 + r'^4 - 2r'^2 \cos \delta}$$

$$= \frac{I_0^2 (1 - r^2)}{1 - 2r^2 \cos \delta + r^4}$$

记强度反射率 $R = r^2$

$$= \frac{I_0 (1 - R)}{1 - 2R \cos \delta + R^2} = \frac{I_0 (1 - R)^2}{(R - 1)^2 + 2R (1 - \cos \delta)} = \frac{I_0}{1 + \frac{4R \sin^2 \frac{\delta}{2}}{(1 - R)^2}}$$

$$I_R = I_0 - I_T$$

$$4R \sin^2 \frac{\delta}{2}$$

$$(1 - R)^2$$

$$= I_0 \left[1 - \frac{1}{1 + \frac{4R \sin^2 \frac{\delta}{2}}{(1 - R)^2}} \right] = I_0$$

$$1 + \frac{4R \sin^2 \frac{\delta}{2}}{(1 - R)^2}$$

$$= I_0 \frac{4R \sin^2(\frac{\delta}{2})}{(1-R)^2 + 4R \sin^2(\frac{\delta}{2})} = \frac{I_0}{1 + \frac{(1-R)^2}{4R \sin^2(\frac{\delta}{2})}}$$

$$R \ll 1 \text{ 时 } I_T \approx I_0 [1 - 4R \sin^2(\frac{\delta}{2})] = I_0 [1 - 2R(1 - \cos \delta)]$$

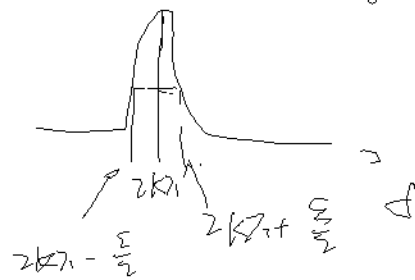
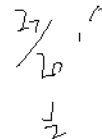
$$I_R = 2I_0 R (1 - \cos \delta)$$

2. F-P干涉装置 半峰宽度

$$I_T = \frac{I_0}{1 + \frac{4R \sin^2(\frac{\delta}{2})}{(1-R)^2}}$$



$$\delta = \begin{cases} 2k\lambda & \text{极大 } I_{T \max} = I_0 \\ (2k+1)\lambda & \text{极小 } I_{T \min} = \frac{1-R}{1+R} I_0 \end{cases}$$



$$\rightarrow \frac{1}{2} = \frac{1}{1 + \frac{4R \sin^2(\frac{\delta}{2})}{(1-R)^2}} \quad \arcsin x \approx x \quad R \rightarrow 1, \delta \rightarrow 0$$

$$\rightarrow \delta = 4 \arcsin \sqrt{\frac{1-R}{4R}} \approx 4 \cdot \frac{1-R}{2\sqrt{R}} = \frac{2(1-R)}{\sqrt{R}}$$

δ 影响因素: 讨论这一点的起因: δ 损失与偏离 2k\lambda 多大时, 亮纹到世界
对应到 $\begin{cases} \delta: \text{角宽度 左算 } \Delta i_k? \text{ 为什么算 } \Delta i_k \\ \lambda \text{ 在 } (\lambda_k - \frac{\delta \lambda_k}{2}, \lambda_k + \frac{\delta \lambda_k}{2}) \end{cases}$

(1) i (单色扩展光入射)

$$\delta, \frac{4\pi n h \cos i}{\lambda} \quad d\delta = - \frac{4\pi n h \sin i}{\lambda} di \quad \text{范围内可以产生}$$

$$\Delta i_k = \frac{\lambda \delta}{4\pi n h \sin i_k} = \frac{\lambda}{2\pi n h \sin i_k} \frac{1-R}{\sqrt{R}}$$

$$n \sin i_k \approx \sin i_{k0}$$

(2) λ (非单色平行光)

$$k \lambda \approx k \lambda_0 \quad C \quad k \lambda$$

$$2nh \cos i = \lambda_k \Rightarrow \nu_k = \frac{c}{\lambda_k}$$

光程差 $\Delta \nu = \frac{c}{2nh}$

对 j 级: $d\delta = -\frac{4\pi nh \cos i}{\lambda^2} d\lambda = \epsilon$

单级线宽 $\Delta \lambda_k = \frac{\lambda_k^2 \epsilon}{4\pi nh \cos i} = \frac{\lambda_k^2}{2\pi nh \cos i} \frac{1-R}{\sqrt{R}} = \frac{\lambda_k}{2k \lambda_k} \frac{1-R}{\sqrt{R}}$

频宽 $\Delta \nu_k = \frac{c}{\lambda_k^2} \Delta \lambda = \frac{c}{\pi k \lambda_k} \frac{1-R}{\sqrt{R}}$

对于波长为 λ , $\lambda + \delta \lambda$ 的波线结构

k 级: $2nh \cos i_k = k \lambda$

$2nh \cos i_k' = k (\lambda + \delta \lambda)$

$\cos i_k' - \cos i_k = -2 \sin \frac{i_k' + i_k}{2} \sin \frac{i_k' - i_k}{2}$

$\approx -\sin i_k \cdot \delta i_k$

$\rightarrow k$ 级条纹的角间距, $\delta i_k = -\frac{k}{2nh \sin i_k} \delta \lambda$

角分辨率, $\delta i_k = \Delta i_k = \frac{\lambda}{2nh \sin i_k} \frac{1-R}{\sqrt{R}}$

$\rightarrow \delta \lambda = \frac{\lambda}{2k} \frac{1-R}{\sqrt{R}}$

色分辨率 $\frac{\lambda}{\delta \lambda} = 2k \frac{\sqrt{R}}{1-R}$